

Contents

Biochemistry and Biophysics	5
A Comparative Study Investigating Human T-Cell Isolation Kits in Multiple Myeloma	5
Graphing the Fold: Simulating the Energy Landscapes of ApoE3 and ApoE4	6
Optimizing Activity Assays for Alpha-Lytic Protease: A Foundation for Studying Protein Folding in <i>Lysobacter enzymogenes</i>	7
Synthesizing Naphthalenediimide-Based Self-Assembling Nanotubes	8
Biology	9
Bird Feeders are Bear Feeders: Studying the Behavior of <i>Ursus americanus</i> in Massachusetts Using Citizen Science	9
Interleukin-6 licenses mast cell regranulation through CD71 upregulation after IgE-mediated activation in mice	10
MYC Super Enhancer Activity in T-ALL Requires TCF-1	11
Phenocopying Δ acrZ through Chromosomal Mutations in the AcrB Efflux Pump	12
Protease-cleavable masking strategy to reduce toxicity of de novo cytokine therapeutics	13
Chemistry	15
Chemical Comparison of Two Compact Cores in the High-Mass Star-Forming Region IRAS 07299-1651	15
Depolymerization of Waste PMMA	16
Development of an Efficient Synthetic Route to Azangulene for the Investigation of Ferroelectric Properties	17
Exploring the Photophysics of Mn ²⁺ Dopant in ZnSe QD System	18
Investigating Localized Surface Plasmon Resonance Effects in Hybrid AuNP-TADF Thin Films	19
Investigating Novel Synthesis of Organic Radical Multiferroics	20
Investigating the Catalytic Activity of ONO- Group V Alkoxide Complexes for the Ring-Opening Polymerization of Biodegradable Materials	21
Kinetics of Solid-State Polymerization of H6TNAP	22
Microwaving Molecules Until They Tell Us Their Secrets: Spectroscopic Characterization of a Halopropene	23
Optimizing Synthesis of InP/MnZnS Quantum Dots	24
Probing the Optical Effects of Azulene Intercalation on a Porous 2D Perovskite, APOSS[MnCl ₄] ₄	25
Synthesis and Evaluation of Novel Alkyl-Sulfonated Near Infrared-II Fluorophores for Cancer Imaging	26
Synthesis of Naphthalenediimide Threonine Derivatives for Nanotube Formation	27
Synthesis of Poly(Carbonate-Ethers) from Organocatalytic, Ring-Opening Polymerization	28
Synthesizing Osteoinductive Graphenic Materials More Efficiently and Cost-Effectively	29
Understanding Artificial Photosynthesis with Quantum Dots and Cobalt Porphyrin Catalysts	30
Computer Science	31
Parallel array characterization and preliminary SOT-MRAM modeling in MemSysExplorer	31

Show, Don't Ask: Resolving ambiguities in interactive Text-to-SQL environments using differentiated databases	32
Mathematics and Statistics	33
An Agent-Based Model of C. difficile Transmission Between Hospitals and Communities	33
Supersymmetric Quantum Mechanics on a Simplicial Complex	34
Medicine	35
Removing Barriers to Alcohol Use Disorder Treatment: Public Interest in Over-the-Counter Naltrexone	35
Neuroscience	37
The Effect of Anesthetic (Tricaine) on Larval Zebrafish Startle Habituation	37
The Role of m6A mRNA Methylation in D2 Striatal Neurons During Pavlovian and Instrumental Learning and Extinction	38
Physics and Astronomy	39
3D Printing Metastructure Legs for High-Efficiency Thermoelectric Generators	39
A Pneumatically Actuated Optical Mount for Enabling Beam Interchange	40
Brane trajectories in cosmological spacetimes	41
Modelling the Twins Detector at the Black Hills Underground Campus (BHUC)	42
Protoplanetary Disk Processing: Centering and Deprojecting	43
SnO ₂ -Coated AFM Tips for Nanoscale Electrical Characterization of Perovskite Solar Cells . . .	44
The Effect of Outer Giant Planets on Inner System Architectures	45
Water-Cooling Acousto-Optics for a Paul Ion Trap	46
Psychology	47
Community Perspectives on Civic Engagement: Findings from Amherst's Civic Academy	47
Sixty Years of Ethnic Studies: A Scoping Review of Emerging Research on Ethnic Studies Policy and Impacts on Adolescent Development	48

Biochemistry and Biophysics

A Comparative Study Investigating Human T-Cell Isolation Kits in Multiple Myeloma

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Purpose. T cells are critical components of the immune response against multiple myeloma (MM), with CD3 serving as a marker of T cells and CD8 identifying cytotoxic T-cell subsets capable of targeting tumor cells. Efficient isolation of T cells is therefore important for studying their anti-tumor activity. This study compared the efficiency of the Miltenyi Biotec and PluriSpin negative T-cell isolation kits and evaluated whether differences in isolation methods affected T-cell functionality against MM tumor cells.

Methods. Human T cells were isolated using either the Miltenyi Biotec or PluriSpin negative T-cell isolation kit. Isolation efficiency and CD3+ T-cell purity were assessed using flow cytometry. Isolated patient T cells were subsequently co-cultured with H929 human MM tumor cells to assess cytotoxic T-cell functionality through a tumor-cell killing assay. The killing potential of T cells isolated using each kit was compared.

Results. Flow cytometric analysis demonstrated greater CD3+ T-cell purity following isolation with the Miltenyi Biotec kit compared with the PluriSpin kit. The Miltenyi isolated T cells also demonstrated greater tumor cell killing potential in the functional killing assay. However, T cells isolated using the PluriSpin kit retained the ability to kill MM tumor cells, indicating that both isolation methods yielded functionally active T cells. Visual assessment of the killing assays did not reveal large differences between the two isolation methods.

Conclusion. The Miltenyi Biotec negative T-cell isolation kit provided improved CD3+ T-cell isolation compared with the PluriSpin kit based on flow cytometric purity assessment. T cells isolated using the Miltenyi kit demonstrated greater cytotoxic potential against MM tumor cells, although PluriSpin-isolated T cells also maintained tumor-cell killing activity. These findings suggest that the choice of T-cell isolation method may influence both the purity and functional assessment of T cells in studies of anti-tumor immunity in multiple myeloma.

Graphing the Fold: Simulating the Energy Landscapes of ApoE3 and ApoE4

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Purpose. Apolipoprotein E (ApoE) is a lipid transport protein in the brain with three isoforms — E2, E3, and E4 — that differ by only one or two residues yet confer strikingly different risks of Alzheimer's disease, with ApoE4 associated with a significantly increased risk. Despite their near-identical sequence and structure, ApoE3 and ApoE4 carry very different disease risk. This project investigates whether differences in protein-folding stability and dynamics can help explain the gap, focusing on the N-terminal fragments (NTFs) of ApoE3 and ApoE4.

Methods. Folding and unfolding kinetics were characterized using hydrogen exchange mass spectrometry (HX-MS), which tracks the exchange of backbone amide hydrogens for deuterium over time as a readout of protein stability. Quantitative folding and unfolding rate constants were extracted from HX-MS data using NumSimEX, a Gillespie-algorithm-based numerical simulation tool, by fitting simulated exchange timecourses to experimental data ($n_p = 5000$ simulated proteins). These rate constants were used to construct free energy landscapes for each isoform.

Results. ApoE3 (NTF) was found to be more thermodynamically stable than ApoE4 (NTF), with a higher free energy of unfolding ($\Delta G = 20.07$ kJ/mol vs. 17.07 kJ/mol, a 3.00 kJ/mol difference). ApoE3 was also more kinetically stable, unfolding roughly 6.7 times more slowly than ApoE4 ($k_u \approx 0.03$ s⁻¹ vs. 0.2 s⁻¹) and facing a higher energy barrier to unfolding ($\Delta G^\ddagger = 81.43$ kJ/mol vs. 76.74 kJ/mol). Notably, ApoE3 protected only about 3% of its exchangeable sites in the native state, compared to roughly 75% for ApoE4, suggesting that ApoE4's native structure is inherently more open and dynamic, which may help explain its reduced stability.

Conclusion. These findings suggest that ApoE3 and ApoE4 differ measurably in both thermodynamic and kinetic folding stability despite their near-identical sequences, and that ApoE4's more open native-state dynamics may contribute to its reduced structural stability. Future work will extend this analysis to a 3-state folding model, address parameter non-identifiability by reporting ranges across multiple fitting seeds, and test NumSimEX's generalizability on additional HX-MS datasets beyond ApoE3/ApoE4.

Optimizing Activity Assays for Alpha-Lytic Protease: A Foundation for Studying Protein Folding in *Lysobacter enzymogenes*

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Purpose. Purpose: Alpha-lytic protease (α LP) and beta-lytic metalloendopeptidase (β LM) are two enzymes produced by the soil bacteria *Lysobacter enzymogenes*. Its unique folding process makes both α LP and β LM interesting for industrial and agricultural use. The goal of our study is to develop a method for measuring α LP activity.

Methods. Methods: α LP activity was measured based on the breakdown of a synthetic substrate (Suc-Ala-Ala-Pro-Phe-pNA). Two measurement techniques were used: a UV/Vis spectrophotometer and a 96-well plate reader, they were compared for their sensitivity, sample volume requirements, and efficiency using serial dilutions of an archived α LP stock (2015). In addition to stock, α LP *L. enzymogenes* was cultured. Its growth was studied in two different types of nutrient media (Bachovchin's Media and Tryptone Soya Broth) to evaluate bacterial growth and natural α LP secretion, monitored by optical density and activity assays.

Results. Results: The plate reader produced clean and linear results. It required less volume of sample while allowing for multiple dilutions to be tested simultaneously, making it the more practical method going forward. A 1:100 dilution of the α LP stock yielded the most favorable and consistent readings. As for the cultivated *L. enzymogenes* bacterial growth differed between the two media, with lower optical density observed in Tryptone Soya Broth.

Conclusion. Conclusion: This study establishes a more efficient, plate reader-based protocol for measuring α LP activity and identifies key next steps for improving bacterial culture conditions to support future studies. These findings lay the groundwork for extending this method to β LM and for exploring alternative expression systems, such as *Bacillus subtilis*, ultimately enabling comparative research into how these two related proteases fold and stabilize.

Synthesizing Naphthalenediimide-Based Self-Assembling Nanotubes

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Purpose. With an emphasis on supramolecular chemistry, this summer we further investigated the serendipitous discovery in Sanders' lab: amino acid NDI (Naphthalenediimide) derivatives that form chiral helical nanotubes through reversible noncovalent interactions. Post DCM (dichloromethane) ban, our main focus has been finding an alternative solvent system to continue experimenting with different linkers for dimer and trimer formation. These iterations of NDI linkages have been shown to increase nanotube stability in previous Hansen lab research.

Methods. Organic chemical synthesis typically involves chemical reactions, purification, and analysis. We utilized microwave chemistry to selectively add L-phenylalanine-t-butyl ester and L-serine-t-butyl ester to the NDI core following the Sander's lab technique. We also used flash chromatography to isolate the target compounds and NMR spectroscopy to confirm the synthesis of the target product.

Results. After testing various solvents, including 1,4-dioxane and acetonitrile, the latter appeared to be the most effective. We continued incorporating different solvophobic linkers to the proposed dimer. Providentially, we also discovered that using a 10-fold excess of diacyl chloride linker led to a more efficient monocoupling of the monomer control molecule, bypassing the previous need for complex organometallic chemistry in this synthesis.

Conclusion. Future thesis work will focus on the trimer synthesis, utilizing similar methods and analysis. Alternative solvents would also need to be considered for the final steps of the synthesis, as part of a potential investigation into linker stability for nanotube formation. Given the ease of synthesizing and customizing NDI derivatives via microwave chemistry, this technique is very promising for continued research on synthetic self-assembling nanotubes for potential applications in drug delivery systems and catalytic reactions.

Biology

Bird Feeders are Bear Feeders: Studying the Behavior of *Ursus americanus* in Massachusetts Using Citizen Science

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Purpose. Increasing urbanization leads to an increase in human-wildlife interaction. One way to monitor wildlife behavior in human-occupied areas is through citizen science, which involves community members submitting data. Here, we sought to explore what behaviors bears (*Ursus americanus*) were exhibiting and what types of behaviors people tended to report in Massachusetts. Observations may be biased positively, as citizen science is often motivated by interest in conservation. Alternatively, people may be likely to report negative interactions due to fear or negative attitudes towards bears.

Methods. Community volunteers submitted photos of black bears in Massachusetts through the MassBears website. We used photos submitted between April 16, 2019 and July 6, 2026. We reviewed photos to produce a coding scheme, similar to an ethogram, which we then used to define observations and behaviors in each photo. Using R, we assessed the proportion of bears exhibiting each type of behavior. Further, we mapped locations of bears in ArcGISPro.

Results. Volunteers submitted 1179 photos. Of these photos, 92.7% contained bears, 6.1% had evidence of bears, and 1.2% had damage caused by bears. For bear photos, 15.3% were either eating in the photo or the volunteer indicated that they were eating. Of submissions showing the bear eating or with the volunteer indicating that the bear was eating, 52.5% of submissions indicated the bear was eating bird food. Most sightings of bears were submitted during the summer, whereas the fewest sightings were submitted during the winter. Bears similarly were observed eating bird food most often in summer months. Of the 1,093 bear photos, 117 showed a bear eating.

Conclusion. The vast majority of sightings were bears, rather than evidence of bears or damage caused by bears. This suggests that fear or negative feelings about bears were not the motivating factor in reporting sightings. It is possible that these types of sightings indicate a more positive attitude towards bears. However, this necessitates further investigation. Gaining a better understanding of the behaviors of black bears in Massachusetts can assist in reducing conflict and fostering human-wildlife coexistence.

Interleukin-6 licenses mast cell regranulation through CD71 upregulation after IgE-mediated activation in mice

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Purpose. Mast cells (MCs) are immune cells that contribute to allergic inflammation by releasing mediators including histamine, tryptase, and leukotrienes upon activation. After immunoglobulin E (IgE)-mediated degranulation, MCs can recover and replenish their secretory granules through regranulation. We previously identified that CD71 expression in MCs was upregulated following IgE-driven activation. Therefore, this study investigated how CD71, the transferrin receptor for cellular iron uptake, is regulated during MC recovery after IgE activation.

Methods. Bone marrow cells were cultured with 30 ng/mL stem cell factor (SCF) and 10 ng/mL interleukin (IL)-3 for 4–6 weeks to generate bone marrow-derived cultured MCs (BMMCs). Cells were sensitized with 1 µg/mL anti-dinitrophenyl-bovine serum albumin (DNP)-IgE overnight and activated with 0.2 µg/mL DNP-BSA. Degranulation was assessed 1 hour after activation, and regranulation was assessed at 1 and 7 days by toluidine blue staining. CD71 expression during recovery was measured by flow cytometry with or without IL-6 treatment. Chromatin immunoprecipitation sequencing (ChIP-seq) datasets were analyzed to identify IL-6-associated transcription factor binding near the mouse CD71 locus. CRISPR-Cas9-mediated gene editing was used to disrupt CD71 in BMMCs, followed by assessment of viability and regranulation.

Results. BMMCs retained the ability to regranulate following IgE-mediated activation. Degranulation was evident 1 hour after activation and granules were restored by day 7. ChIP-seq analysis identified binding sites for multiple IL-6-associated transcription factors, including signal transducer and activator of transcription (STAT) 1, STAT3, STAT5A, STAT5B, JUN, ELK1, and CCAAT/enhancer-binding protein beta (C/EBPβ), overlapping an active enhancer region near the mouse CD71 locus. CD71 expression increased following activation and gradually declined over time, but remained elevated through day 7 with IL-6 treatment. CRISPR-Cas9-mediated disruption of CD71 did not substantially affect BMMC viability but impaired restoration of granules after IgE-mediated activation compared with untransfected and off-target controls.

Conclusion. These findings identify IL-6 as a critical regulator of MC regranulation following IgE-mediated activation by promoting CD71 expression. IL-6-associated transcriptional regulation may promote CD71 expression during recovery, potentially supporting iron-dependent processes required to restore MC granules. This pathway may help explain how MCs recover their secretory capacity following activation and remain capable of subsequent inflammatory responses.

MYC Super Enhancer Activity in T-ALL Requires TCF-1

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Purpose. Aberrant activation of MYC is a common oncogenic event across human cancers, including T-cell acute lymphoblastic leukemia (T-ALL). In approximately 5% of human T-ALL cases, MYC overexpression results from duplication and activation of N-ME, a distal Myc super-enhancer. We have developed a spontaneous mouse model of T-ALL driven by T-cell-specific stabilization of β -catenin combined with the loss of the transcription factor HEB (CATHEB Δ). This model faithfully recapitulates spontaneous N-ME super enhancer duplication and MYC overexpression during the early double positive (DP blast) stage of thymocyte development. We have also identified novel recruitment of TCF-1 to the N-ME super enhancer, and genetic ablation of TCF-1 from CATHEB Δ thymocytes completely prevents leukemogenesis. However, it remained unclear whether this protection resulted from disruption of thymocytes development, or whether TCF-1 has a key role in N-ME superenhancer activation via chromatin accessibility and remodeling.

Methods. To distinguish between these possibilities we characterized thymocytes development in CATHEB Δ TCF1 Δ mice using multiparameter flow cytometry with CD4, CD8, CD69, CCR7, CD25, TCR β , TCF-1, and viability staining.

Results. Although total thymocytes numbers were significantly reduced consistent with the established role of TCF-1 in thymocyte development, CATHEB Δ TCF1 Δ thymocytes progressed to the DP blast stage that is targeted for leukemogenesis in CATHEB Δ mice. In addition, these mice exhibited significantly increased proportions of mature, post-DP T cells expressing CD69+ and CCR7+.

Conclusion. These findings demonstrate that loss of leukemogenesis in CATHEB Δ TCF1 Δ mice cannot be explained solely by a developmental block at the DP stage. Instead, they support a direct role for TCF-1 in promoting leukemic transformation, consistent with its proposed function in activating the N-Me super-enhancer and driving MYC-dependent T-ALL.

Phenocopying Δ acrZ through Chromosomal Mutations in the AcrB Efflux Pump

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Purpose. Antibiotics are essential in treating bacterial infections. However, some bacteria have developed the ability to survive in the presence of antibiotics. This antibiotic resistance is a threat to individual and public health. In *Escherichia coli*, one such mechanism is the AcrAB-TolC multidrug efflux pump, a protein pump in the bacteria's cell membrane that pumps antibiotics out of the cell. The deletion of the small protein AcrZ, which is associated with the pump, changes susceptibility to some antibiotics. To better understand the specific role of AcrZ, its relation to AcrB, and its interaction with certain classes of antibiotics, we sought to phenocopy the impact of an *acrZ* deletion through point mutations corresponding to residues 563 and 676 in the amino acid sequence for AcrB.

Methods. We first inserted the *cat-sacB* cassette, which confers chloramphenicol resistance and sucrose sensitivity, onto the *E. coli* chromosome using lambda red recombineering and allelic exchange. The cassette allowed us to test for proper chromosomal insertion using the *cat-sacB* phenotype, with the goal of later replacing *cat-sacB* with our mutants of interest.

Results. While both allelic exchange and lambda red recombineering yielded chloramphenicol resistant strains, neither method produced sucrose-sensitive strains, indicating improper integration of *cat-sacB* or *sacB* mutation.

Conclusion. Further troubleshooting and sequencing of the strains are required before our mutants of interest can be inserted.

Protease-cleavable masking strategy to reduce toxicity of de novo cytokine therapeutics

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Purpose. Cytokines are immune signaling proteins that can induce antitumor responses by stimulating cytotoxic lymphocytes (CTLs). However, clinical adoption of cytokine immunotherapies has been hindered by systemic immune activation and dose-limiting toxicity. We investigated a de novo protein-design strategy for synthesizing protease-activated cytokine prodrugs. These prodrugs consist of a therapeutic cytokine bound to a masking domain via a linker cleaved by tumor-associated matrix metalloproteinases (MMPs). In its masked state, the cytokine cannot associate with its cognate receptor and remains inactive. Once the linker is cleaved by MMPs, the cytokine regains pharmacological activity and stimulates tumor-targeting CTLs. We reasoned that conditional prodrug activation in tumors can limit systemic toxicity while retaining potent anti-tumor efficacy.

Methods. We cultured a KPC cell line derived from murine pancreatic ductal adenocarcinoma (PDAC) and injected the cancer cells subcutaneously into 20 healthy wild-type mice. By day 13, the mice had developed palpable tumors and were tail-vein injected with 21h10 or Neo-2 treatments. Four treatment conditions were used for both cytokines: untreated (negative control), unmasked cytokine (positive control), masked cytokine (experimental group), and a masked construct with a non-cleavable linker (negative control). Three hours after injection, the mice were euthanized, and their spleens and tumors were harvested and lysed in RIPA lysis buffer. Western blot densitometric analysis was used to measure the ratio of phosphorylated STAT (P-STAT) to total STAT, quantified as fold change relative to untreated controls, using ImageJ and GraphPad Prism.

Results. In Neo-2-treated mice, masked Neo-2 induced negligible P-STAT5 signaling in the spleen but higher levels in the tumor, consistent with preferential tumor activation. In contrast, unmasked Neo-2 produced low P-STAT5 levels in both tissues, comparable to untreated controls. Similarly, masked 21h10 induced negligible P-STAT3 signaling in the spleen but higher levels in the tumor. Unmasked 21h10 produced low P-STAT3 levels in both tissues, comparable to untreated controls.

Conclusion. Our results suggest that masking enables preferential activation of cytokine prodrugs within tumors while limiting systemic signaling. Negligible STAT phosphorylation in the spleen following masked cytokine treatment, together with increased tumor signaling, supports the potential of this approach to reduce off-target immune activation and toxicity. However, further in vivo studies in tumor-bearing mice are needed to determine whether preferential tumor activation translates into improved therapeutic outcomes. Future

experiments should assess anti-tumor efficacy through tumor growth and survival, and systemic toxicity through changes in body weight.

Chemistry

Chemical Comparison of Two Compact Cores in the High-Mass Star-Forming Region IRAS 07299-1651

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Purpose. High-mass star-forming regions contain warm, dense gas with rich molecular emission, but neighboring protostellar cores may experience different chemical and physical conditions. This study compares two compact continuum cores in the Infrared Astronomical Satellite (IRAS) source 07299-1651 to determine whether their molecular inventories and kinematic properties are similar or chemically differentiated.

Methods. Submillimeter Array (SMA) continuum maps and spectral cubes were analyzed across 24 spectral windows. Astrodendro was used to identify the two continuum peaks, and their pixel positions were used to extract spectra with Common Astronomy Software Applications (CASA). Emission features were counted, and representative profiles were compared using one- and two-component Gaussian models. The eXtended CASA Line Analysis Software Suite (XCLASS) generated synthetic spectra and fitted molecular column densities, velocities, and linewidths for each core.

Results. XCLASS assigned molecular carriers to 96% and 94% of the counted emission features in Cores 1 and 2, respectively. The cores display broadly similar molecular inventories, but their fitted column densities vary by species. Several molecules have higher fitted column densities toward Core 2, whereas sulfur monoxide and the cyano radical are enhanced toward Core 1. Mean fitted velocities of 17.1 km/s for Core 1 and 16.8 km/s for Core 2 are close to the reported systemic velocity of 16.6 km/s. Core 2 has a broader mean molecular linewidth than Core 1, 6.1 versus 3.6 km/s. A two-component Gaussian model better reproduces a representative line. The model is favored over a single-component model, indicating that blended or unresolved velocity components may contribute to the observed profiles.

Conclusion. The two compact cores in IRAS 07299-1651 are chemically related but not identical. Species-dependent column densities and the broader linewidth of Core 2 indicate differences in their local chemical or kinematic environments, while their mean velocities remain consistent with the surrounding region. Future work will refine uncertain molecular fits and apply the analysis pipeline to nine additional high- and intermediate-mass star-forming regions observed with the SMA.

Depolymerization of Waste PMMA

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Purpose. Poly(methyl methacrylate) (PMMA) is widely used for its mechanical properties, but high thermodynamic and kinetic barriers make depolymerization difficult, leading to inefficient waste recycling. Recent work has been done on pre-made polymers but this study aims to modify pre-existing PMMA waste. By incorporating phthalimide ester pendant groups our goal is to achieve high percentages of PMMA depolymerization at lower temperatures.

Methods. Our approach begins with partial hydrolysis (<15%) of colored PMMA waste to install pendent carboxyl groups. Afterwards esterification with N-hydroxyphthalimide to create phthalimide-ester containing PMMA. The phthalimide-ester groups provide thermally sensitive triggers that allow for effective depolymerization at lower temperatures. Nuclear Magnetic Resonance (¹H NMR) spectra was used to confirm conversion of pendent groups and Thermogravimetric Analysis (TGA) was used to determine extent of depolymerization.

Results. ¹H NMR spectra suggests hydrolysis of colored PMMA and subsequent conversion to N-hydroxyphthalimide ester. TGA ramp suggested that depolymerization onset decreased to 240°C significantly lower than PMMA. Isothermal hold showed 85% mass loss, equivalent to 85% depolymerization. Successful post-polymerization modification of P(MMA-co-MAA) and high extents of depolymerization of PMMA were achieved.

Conclusion. We showed that our approach is effective for depolymerizing real world PMMA waste and helps contribute to a circular polymer economy. Future work will focus on expanding depolymerization to other vinyl polymers and post-automotive waste. More work should also be done to scale depolymerization toward techno-economic viability.

Development of an Efficient Synthetic Route to Azangulene for the Investigation of Ferroelectric Properties

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Purpose. We aimed to develop an efficient synthetic route to azangulene, a nitrogen-containing helicene to investigate ferroelectric properties. A previously developed synthetic route had been investigated by the laboratory for several years but consistently encountered difficulties in the mid stages of the synthesis. Further investigation identified a fundamental reactivity conflict: conditions that promoted a high-yielding reaction also promoted undesired debromination, preventing the necessary functional groups from coexisting under the same reaction conditions. Therefore, a modified synthetic route was developed to overcome this limitation.

Methods. The revised route combines elements of the previous azangulene synthesis with strategies from the reported synthesis of borangulene. The first step uses a nucleophilic aromatic substitution under basic conditions to construct the extended aromatic precursor. The resulting nitro compound is then pushed through a reduction using hydrazine hydrate and Pd/C in ethanol at >100 °C. Reaction products were monitored and purified using standard organic synthesis techniques.

Results. Investigation of the previous synthetic route demonstrated that high yield could not be achieved without high probability of debromination, making the original strategy unsuitable for reliably producing the desired intermediate. An alternative route was therefore developed by combining the previous azangulene strategy with elements of the borangulene synthesis. The first two steps of the revised route have been successfully carried out. Including formation of the nitro precursor and its subsequent reduction to the corresponding amine. However, the reduction step is still an area requiring further optimization to obtain consistently high yields with less time spent.

Conclusion. Identification of the competing debromination pathway provided an explanation from the limitations of the previous azangulene synthesis and motivated the development of a revised synthetic route. The successful completion of the first two steps demonstrates initial progress toward the new route. Future work will focus on optimizing the reduction step and determining suitable conditions for the remaining steps. Successful completion of the revised synthesis would provide a more efficient route to azangulene for the following investigation of its ferroelectric properties.

Exploring the Photophysics of Mn²⁺ Dopant in ZnSe QD System

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Purpose. To learn more about the Mn:ZnSe system, we aim to produce a series of Mn:ZnSe QDs with different doping amounts to observe their characteristics and differences. We also want to adjust the synthesis procedure to tune the size and produce uniform dots. We aim to also optimize the EPR settings as Mn has a very strong signal. To further understand the spin of Mn²⁺, we will do CW-EPR to characterize the spin of Mn²⁺ at varying concentrations in ZnSe QDs.

Methods. Mn:ZnSe quantum dots are synthesized in an air free environment, with the degassing and heating times modified to tune QD size. UV-Vis spectroscopy and emission are used to observe absorption, emission, and relative quantum yield. Electron paramagnetic resonance (EPR) spectroscopy is performed to observe the spin of Mn²⁺. To determine size and doping percent of the QDs, transmission electron microscopy and ICP mass spectroscopy are performed.

Results. The results of this study suggest that the Mn:ZnSe QDs synthesized express similar characteristics to those in literature. Samples with higher inputs of Mn have lower lifetimes and higher emission values, as expected. The EPR spectra of samples also display the six peaks expected. However, with very high inputs of Mn, samples experience signal disruption, leading to lower emission, or Mn-Mn coupling, leading to a different EPR signal.

Conclusion. The observations from this study demonstrate potential in synthesizing a broader series of Mn:ZnSe QDs with different Mn²⁺ doping percentages. Beyond just characterization, we also aim to determine the electron transfer rate, through methods such as quenching or shelling, and photophysics of Mn:ZnSe QDs.

Investigating Localized Surface Plasmon Resonance Effects in Hybrid AuNP-TADF Thin Films

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Purpose. Thermally activated delayed fluorescence (TADF) materials exhibit temperature-dependent photophysical changes, as thermal energy facilitates reverse intersystem crossing from the triplet to the singlet excited states, giving rise to delayed fluorescence. Gold nanoparticles (AuNPs) exhibit localized surface plasmon resonance (LSPR), through which absorbed light can generate localized thermal effects. This study investigates how LSPR from AuNPs influences the photophysical behavior of the TADF emitter 2,4,5,6-tetra(9H-carbazol-9-yl)isophthalonitrile (4CzIPN) in polymer thin films, with emphasis on changes in its delayed fluorescence lifetime and its potential as a probe of local temperature changes near AuNPs.

Methods. Polymer thin films containing poly(methyl methacrylate) (PMMA), 4CzIPN, and AuNPs were prepared by spin coating and drop casting and characterized using ultraviolet-visible (UV-Vis) absorption and steady-state emission spectroscopy. Temperature-dependent fluorescence lifetime measurements were used to determine how the lifetime of 4CzIPN changes with temperature. Hybrid AuNP-4CzIPN films were further examined using dark-field microscopy, fluorescence lifetime imaging microscopy (FLIM), and transient absorption spectroscopy to investigate their photophysical response to plasmon excitation.

Results. The delayed fluorescence lifetime of 4CzIPN in polymer thin films decreased with increasing temperature, establishing a relationship between the photophysical response of 4CzIPN and thermal changes in its environment. In hybrid thin films containing both 4CzIPN and AuNPs, a decrease in fluorescence lifetime was also observed following plasmon excitation, consistent with a possible contribution from LSPR-generated heating to the observed lifetime decrease. However, the observed changes in lifetime cannot yet be attributed exclusively to plasmonic heating, as other interactions between AuNPs and 4CzIPN, including quenching, may also contribute to the measured response.

Conclusion. These results show the potential for using the temperature-dependent delayed fluorescence lifetime of 4CzIPN to probe local temperature changes in hybrid AuNP-TADF thin films. Further work is needed to separate LSPR-generated heating from other AuNP-4CzIPN interactions, such as quenching, and determine whether TADF emitters can be used as local temperature probes near plasmonic nanoparticles.

Investigating Novel Synthesis of Organic Radical Multiferroics

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Purpose. Previous research in the Wiscons Lab has explored the peculiar characteristics of "triangulenes," or heteroatom-centered, triply oxygen-bridged triphenyl structures, seeking to work towards organic data storage. This study seeks a new route of synthesis for a carbon radical-centered triangulene that, due to its unpaired electron, should possess an electric and magnetic dipole. A material with both of these qualities is known as "multiferroic," and organic multiferroics are promising candidates for highly manipulable materials.

Methods. To synthesize precursor 1,3-Bis-(3-methoxyphenoxy)benzene, we attempted a copper-catalyzed Ullmann coupling reaction to combine reagents resorcinol and 3-iodoanisole. This reaction ran overnight for 20 hours at 135°C. After reacting, the product was first pulled through a celite plug, then washed with 0.5M HCl in a separatory funnel. The organic layer was then dried with magnesium sulfate, and the product extracted via column chromatography.

Results. Though the scale of the reaction is small, we achieved a relatively high yield; ¹H NMR testing confirmed the product's identity and high degree of purity. 3.1 grams of precursor were successfully synthesized over 8 reactions.

Conclusion. We have accrued enough precursor to proceed to the next steps: lithiation and ring-closing. Because we cannot safely work with n-butyllithium ourselves, Dr. Brian H. Northrop and his group at Wesleyan University have graciously agreed to perform the lithiation and nucleophilic ring-closing steps. We hope that this will yield a triangulene cation that can be subsequently reduced to a radical in future work.

Investigating the Catalytic Activity of ONO- Group V Alkoxide Complexes for the Ring-Opening Polymerization of Biodegradable Materials

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Purpose. We are looking to expand the overlooked usage of Group V metals in the creation of catalysts that use ring-opening polymerization to synthesize biodegradable plastics.

Methods. Using various commercially available compounds we synthesized a variety of unique ligands based on previous students' studies and relevant literature. These ligands, produced under fume hoods in a normal atmosphere, were then transferred into an oxygen and water free glovebox and used to create catalysts. We then observed the structures of these chemicals with Nuclear Magnetic Resonance (NMR) and Single Crystal X-Ray Diffraction (SCXRD) technology. Once made, these catalysts were combined with various monomers in the glovebox and melted together at 140°C so their polymerization could be analyzed.

Results. We were able to observe the changes to the properties of catalysts when made with Niobium versus Tantalum as well as how different functional groups present on the ligands used affected our catalysts. We crystalized some of these catalysts and, using the SCXRD, were able to view their real structures and how they compare to our NMR spectra. By viewing the kinetics data from our polymerizations, we determined the different rates at which the catalysts are able to operate.

Conclusion. By knowing how a variety of functional groups on our ligands changes the efficacy of our catalysts, we will be better able to isolate the most effective groups and in turn make better catalysts with these Group V metals. As we make better catalysts, the biodegradable plastics we can produce will also improve and their uses in society can continue to expand.

Kinetics of Solid-State Polymerization of H6TNAP

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Purpose. 11,11,12,12-Tetracyanonaphtho-2,6-quinodimethane, which we abbreviate as H6TNAP, is one of few known molecules to polymerize in the solid state. Unlike polymerization in the liquid or gas state, solid state polymerization provides a high degree of control over polymerization reactions. These controlled polymer syntheses have applications in materials science, specifically for producing copolymers with the capability to stabilize donor-acceptor interfaces in electronic materials, like semiconductors and photovolatics. However, due to the rarity of solid-state polymerization in organic materials, much is still unknown about their kinetics. Our project seeks to explore the kinetics of solid-state polymerization in an organic crystalline substance, focusing on how the polymerization propagates and the factors that explain why.

Methods. H6TNAP Polymerization occurs under conditions of high heat and UV-Vis light. We created these conditions using a hot plate set between 200-220°C and a photoreactor, 3D printed to keep an LED light ~2 cm away from the sample. We created samples on 40 mm glass cover slips, isolating 5-7 H6TNAP crystals on the center of each and sealing with another cover slip kept 0.06mm apart using one layer of Kapton tape. Because the H6TNAP monomer is red and its polymer is colorless, we collected data using a brightfield microscope, which accurately indicated the polymerization using average intensity measurements. Images were captured every 5 minutes and analyzed using Zen Imaging software to create time series for each trial.

Results. We observed both linear and quadratic average intensity time series. Linear time series were most often observed in data sets with less time points, while quadratic time series were observed in longer data sets, like those with larger crystals or lower temperature trials. Via image analysis, we observed that perimeter to area ratio increased over time and that core intensity increased over time as well.

Conclusion. Given that shorter time series yielded linear data, we hypothesized that the standard polymerization kinetics of our organic crystals are linear by area. Under this assumption, our image analysis observations allowed us to identify factors that explained the quadratic average intensity time series, namely an increasing perimeter:area ratio and 3D polymerization, which we hypothesized to cause slight increases in polymerization rate over time in larger crystals or lower temperature trials. Currently, we do not have adequate data to necessarily prove our hypotheses. Our next steps are to improve methods as to run trials more efficiently and with less miscellaneous error.

Microwaving Molecules Until They Tell Us Their Secrets: Spectroscopic Characterization of a Halopropene

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Purpose. In quantum systems such as molecules and their complexes, angular momentum is quantized. This quantization yields distinct transitions that can be measured and exploited to determine the precise structure of the molecule and its complexes with other atoms. By determining the structure of such a system, a fuller understanding of intermolecular forces can be advanced. In this project, we use microwave spectroscopy to characterize the compound 3-chloro-1,1,3,3-tetrafluoropropene and its complex with argon.

Methods. An experimental spectrum was measured using Chirped-Pulse Fourier-Transform Rotational Microwave (CP-FTMW) Spectroscopy. To predict discrete transitions, the system was modeled with the quantum chemistry software Gaussian16. Subsequently, the theoretical spectrum was simulated and assigned to the experimental spectrum using PGOPHER, and revised structural coordinates were calculated where possible.

Results. A dihedral scan of the monomer revealed two enantiomeric rotamers, which have identical spectroscopic constants. We then matched peaks for the Cl35 and Cl37 isotopologues of the monomer. We fit rotational constants, distortion constants, and quadrupole coupling constants to the best of their ability based on the transitions assigned. We then scanned Argon around the monomer for the argon complex, and we took the lowest energy configuration to assign transitions for the Cl35 and Cl37 isotopologues of the argon complex.

Conclusion. The theory we used provided an estimate that lay within 10% of the value of our assigned constants, producing a relatively accurate theoretical structure. In the future, we want to look at the acetylene complex with the monomer to find out more about the structure of the molecule and its intermolecular interactions.

Optimizing Synthesis of InP/MnZnS Quantum Dots

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Purpose. This study aimed to synthesize and characterize manganese (Mn)-doped core/shell indium phosphide/zinc sulfide (InP/ZnS) quantum dots (QDs) with blue and green emission. Mn doping introduces additional optical properties to semiconductor QDs, making these materials of interest for applications in optoelectronics. A major challenge encountered during synthesis was the formation of phosphate byproducts, which interfered with the characterization of the desired QDs. We focused on modifying the synthesis conditions to improve QD formation while minimizing the amount of phosphates.

Methods. InP/ZnS QDs were synthesized using a hot-injection method, with modifications made to reaction conditions between syntheses to optimize QD formation. Distinct samples were synthesized in the green-emitting series. Absorption and emission spectroscopy were used to characterize the QDs and determine their emission behavior, while transmission electron microscopy (TEM) was used to examine the morphology and size of QDs. Changes made between syntheses were assessed based on their effects on product formation and characterization.

Results. Sample MG02 served as a control sample for comparing subsequent modifications and showed significant phosphate formation. Repeating the synthesis with a degassed petrolatum jelly (PJ):octadecene (ODE) mixture and increased stirring speed in sample MG03 did not substantially resolve phosphate formation. Changing the inert atmosphere from house nitrogen to Grade 5 argon in sample MG04 resulted in a decrease in the amount of phosphate observed. Further modifications in sample JHK22, based on previously successful synthesis protocols, including consolidation of the precursor injections, use of a fresh tris(diethylamino)phosphine source, and stricter control of heating times, resulted in a further decrease in phosphate formation, although phosphate byproducts remained present.

Conclusion. Synthesis modifications for the InP/ZnS QDs procedure demonstrated that reaction conditions, including inert atmosphere, precursor amount, injection strategy, and heating control, can influence phosphate byproduct formation. The decrease in phosphate observed following the use of Grade 5 argon and subsequent protocol modifications suggests that these changes improved the synthesis conditions, although further optimization is necessary to eliminate phosphate formation and obtain consistently characterizable QDs. Uv-Vis/photoluminescence spectroscopy and TEM characterization provided methods for identifying conditions of successful Mn-doped and undoped InP/ZnS QDs.

Probing the Optical Effects of Azulene Intercalation on a Porous 2D Perovskite, APOSS[MnCl₄]₄

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Purpose. Manganese (Mn)-based perovskites, such as porous APOSS[MnCl₄]₄, are excellent light emitters, despite being poor absorbers. It has been shown in other Mn-based perovskites that select organics can transfer some of their absorbed energy to the Mn lattice, allowing the perovskite to emit light without direct absorption via Forster Resonance Energy Transfer (FRET). This project examines changes in the photoluminescence emission and excitation spectra of APOSS[MnCl₄]₄ following the intercalation of a luminescent molecule, azulene, into its pores.

Methods. APOSS[MnCl₄]₄ was synthesized. Diffuse reflectance spectroscopy (DRS) was run to determine the primary excitation wavelengths of APOSS[MnCl₄]₄ with and without azulene intercalated. Azulene was dissolved in isopropanol and intercalated into APOSS[MnCl₄]₄. The presence and approximate amount of azulene intercalated was found via nuclear magnetic resonance (NMR). Powder X-ray diffraction (PXRD) confirmed that the APOSS[MnCl₄]₄ crystal structure was retained following azulene intercalation. Photoluminescence (PL) and photoluminescence excitation (PLE) spectroscopy was run to test azulene excitation and emission in perovskite pores and search for evidence of FRET.

Results. PXRD confirmed that APOSS[MnCl₄]₄ was successfully synthesized. DRS found the excitation wavelength of the Mn lattice to be 417 nm. PL at 417 nm on APOSS[MnCl₄]₄ showed peaks at 491 nm and 605 nm, which were determined to be from APOSS-Cl₈ and the Mn-Cl lattice, respectively. Using varying azulene concentrations, approximately 0.5 and 1 equivalents azulene were intercalated into APOSS[MnCl₄]₄. Via DRS, azulene excitation was found to occur primarily at 282 nm. At 282 nm, the PL of the azulene intercalated samples showed large peaks at 376 nm, which were not present in the PL of the plain APOSS[MnCl₄]₄. Likewise, PLE of the azulene intercalated samples at 376 nm showed a new peak at 282 nm. PLE at 482 nm of the 0.5 equivalent azulene intercalated sample showed a novel peak at 282 nm, but no such peak appeared in the PLE at 605 nm. No peak at 605 nm was observed in the 282 nm PL in the azulene intercalated sample.

Conclusion. This experiment showed that a luminescent molecule can be intercalated into APOSS[MnCl₄]₄ and that its loading amount is tuneable and concentration-dependent. Azulene was found to be able to excite and emit in APOSS[MnCl₄]₄ pores. There was evidence of energy transfer between the APOSS-Cl₈ organic and azulene. There was no evidence of FRET between the azulene and Mn lattice.

Synthesis and Evaluation of Novel Alkyl-Sulfonated Near Infrared-II Fluorophores for Cancer Imaging

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Purpose. Fluorescence-Guided Surgery (FGS) has emerged as a sophisticated method of intraoperative molecular imaging that addresses the fundamental limitations of conventional intraoperative assessment (white-light visualization, palpation, and mental “registration” of preoperative imaging) [1]. In FGS, a fluorescent probe is administered systemically prior to surgery and allowed to circulate and accumulate in the tumor over time. The currently available fluorophores in FGS emit and absorb in the NIR-I region, which is characterized by shallow penetration and high autofluorescence. To address these limitations, our lab has developed a class of low-molecular-weight organic fluorophores, 4,4'-quinocyanines (QuCy), that extend emission into the NIR-II region [2]. Despite QuCy's success in achieving high-resolution visualization in murine models, its fluorescence is quenched when dissolved in aqueous media due to the formation of aggregates. For this reason, liposomal encapsulations were made to deliver this fluorophore to mice. To resolve this complication, we hypothesize that adding sulfonate groups to the QuCy head will increase their water solubility and prevent aggregation, thereby eliminating the need for a liposome for delivery. Preliminary data show that the alkyl-sulphonated QuCy (sbcQuCy) has a strong absorbance in NIR-II when dissolved in polar solvents like DMSO and Methanol.

Methods. With the synthesized dye, we carried out a cellular uptake assay, cytotoxicity assay, and intravenous injection in naive and tumor mice.

Results. Evaluation of sbcQuCy revealed efficient tumor uptake, low cytotoxicity, and high-resolution imaging

Conclusion. With these findings, our hope is that sbcQuCy will serve as a clinically translatable NIR-II contrast agent

Synthesis of Naphthalenediimide Threonine Derivatives for Nanotube Formation

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Purpose. Supramolecular chemistry investigates noncovalent forces that promote the self-assembly of larger structures, often to model biological systems such as proteins. As observed by the Sanders group, Naphthalenediimide (NDI) derivatives form nanotube structures through hydrogen bonding. Nanotube stability can then be increased by selecting for amino acid hydrophobicity and by adding hydrophobic covalent linkers.

Methods. Naphthalenediimide derivatives were synthesized using microwave synthesis to unsymmetrically add L-Phenylalanine and L-Threonine to the naphthalene core. Coupling procedures were adapted using EDC coupling and acylation techniques to yield a dimer NDI with a terephthalic ester linker. These reactions were adjusted due to the EPA ban on dichloromethane, with dimethyl carbonate and acetonitrile tested as possible solvent replacements. The resulting compounds were observed and characterized using Nuclear Magnetic Resonance (NMR).

Results. EDC coupling using dimethyl carbonate yielded the greatest amounts of mono-coupled and di-coupled material, but a large amount of starting material remained. The dimer product was identified by the β proton shift upfield in the NMR spectrum. When run at higher temperatures, EDC coupling produced an elimination byproduct in greater yield than the dimer. Attempts at acylation of L-threonine alone were primarily successful in producing the mono-acylated material but were abandoned because TFA, which is used in a subsequent procedure to remove the t-butyl protecting groups, was required.

Conclusion. EDC coupling of the NDI L-threonine derivatives was identified as the most suitable procedure for dimerization. These findings will allow for comparison with the NDI L-serine derivative counterpart to examine the effects of the methyl group on nanotube stability and hydrophobicity. Future directions toward this analysis require a procedure to deprotect the t-butyl protecting groups and CD spectra characterization. Additional tuning towards the covalent linker hydrophobicity can also be conducted using naphthalene and anthracene linkers.

Synthesis of Poly(Carbonate-Ethers) from Organocatalytic, Ring-Opening Polymerization

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Purpose. In the medical field, poly(ethylene Glycol) (PEG) has been widely used as a water-soluble, biocompatible polymer material. However, PEG is nondegradable, leading to bioaccumulation and health hazards. Therefore, researchers have turned to degradable alternatives such as poly(carbonate-ethers) synthesized by organocatalytic, ring-opening polymerization. This work expands upon poly(carbonate-ether) synthesis by investigating the coupling of glycerol carbonate (GC) with small molecules and drugs prior to polymerization, yielding degradable, water-soluble polymers that enable easy and diverse functionalization via esterification. Additionally, we explored the synthesis of poly(carbonate-ethers) into bottlebrush polymers. The high density of functional side chains within bottlebrush architectures provides opportunities for increased drug loading and tunable viscosity, potentially enabling more versatile and effective drug delivery systems.

Methods. GC was coupled with the desired small molecule or drug at room temperature for 24 hours using EDC-HCL and DMAP as coupling reagents. The resulting monomer was subjected to ring-opening copolymerization with ethylene carbonate (EC) (20% GCX: 80% EC) at 160 °C for 4 hours using TBD as an organocatalyst and benzyl alcohol as an initiator. For the bottlebrush polymers, norbornene-initiated macromonomers were subjected to ring-opening metathesis polymerization (ROMP) using a Grubbs 3rd-generation catalyst overnight. All structures were confirmed by ¹H NMR and, when suitable crystals formed, by SCXRD. All reactions were handled and synthesized in an anhydrous, O₂-free glovebox.

Results. We successfully coupled GC with BA to form GCBA, and with naproxen (N) to form GCN, as evidenced by ¹H NMR and, when relevant, SCXRD. In addition, ¹H NMR supports the successful copolymerization of GCBA and GCN with EC to form poly(EC-co-EO-co-GCBA-co-GOBA) and poly(EC-co-EO-co-GCN-co-GON), respectively. Finally, a norbornene-initiated GCBA macromonomer was successfully synthesized, as shown by ¹H NMR. However, no bottlebrush polymers were successfully synthesized.

Conclusion. We examined the functionalization of GC with small molecules and drugs prior to copolymerization with EC. Additionally, we explored the synthesis of norbornene-initiated poly(carbonate-ether) bottlebrush polymers. Together, functionalized poly(carbonate-ethers) and their bottlebrush derivatives provide a promising platform for developing biodegradable, water-soluble, and highly functionalizable alternatives to PEG for biomedical applications.

Synthesizing Osteoinductive Graphenic Materials More Efficiently and Cost-Effectively

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Purpose. Currently, metal implants are frequently used to treat serious bone injuries; however, implants do not match the natural composition or porous structure of healthy bone. Because bone adapts to the loads placed under it, metal implants can absorb the load instead of the bone tissue, leading to bone weakening over time. This research project looks at a novel, more efficient, and cost-effective method for the synthesis of calcium phosphate graphene (CaPG), an osteoinductive graphenic material that can encourage bone regeneration.

Methods. The project examined a novel, aqueous synthetic method for making CaPG intermediates (pE/pK-CDI-GO). Graphene Oxide (GO) was synthesized from graphite using modified Hummer's method (using sulfuric acid, sodium nitrate, potassium permanganate, followed by hydrogen peroxide and water), which was then activated with 1,1'-Carbonyldiimidazole (CDI) in water or tetrahydrofuran (THF). Polymer attachment of protected poly-lysine or protected poly-glutamic acid was performed under aqueous conditions. Data on surface functionalization was obtained using X-ray photoelectron spectroscopy (XPS) elemental analysis, with the nitrogen atom percentages measured prior to and following the polymer attachment steps in the aforementioned solvents.

Results. Nitrogen atom percentages following polymer attachment were lower when compared to the control samples (GO after activation with CDI in the two solvents). For the water control, the nitrogen percentage was ~2.1%; following polymer attachment, nitrogen percentage was ~1.3% and 0% for poly-lysine and poly-glutamic acid respectively. For the THF control, the nitrogen percentage was ~2.0%; following polymer attachment, nitrogen percentage was ~0.4% and ~1.5% for poly-lysine and poly-glutamic acid respectively. These findings indicate that the reaction did not proceed as anticipated and significant polymer attachment did not occur.

Conclusion. While the new synthetic pathway offers theoretical benefits in cost, efficiency, and safety compared to previous methods, the experimental data show that significant polymer attachment was not achieved. Future steps involving deprotected polymers may yield better results, as they have more reactive sites.

Understanding Artificial Photosynthesis with Quantum Dots and Cobalt Porphyrin Catalysts

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Purpose. This study aims to understand the photophysics behind the relative performance of several cobalt porphyrin catalysts in quantum dot photocatalysis. Co-H, Co-OMe, Co-CF₃, and Co-COOH perform differently in carbon reduction photocatalysis, and also affect electron transfer in a variety of ways. The goal was to investigate whether the differences in electron transfer could explain the photocatalytic performance.

Methods. In this study, quenching experiments were performed on Zinc Selenide Quantum Dots with all four aforementioned catalysts. The change in emission, absorption, and excitation lifetime was observed carefully. Then, catalysts were used in a quantum dot photocatalysis system and carbon reduction was measured using a gas chromatograph. These data were examined in parallel.

Results. The quenching experiments showed Co-COOH to have the largest impact on electron transfer, as measured through lifetime and emission. Similarly, Co-COOH had the best carbon reduction performance.

Conclusion. More data is needed to confirm that the electron transfer has a causal effect on the photocatalysis system. However, if it were proven, this could demonstrate that the electron transfer speed is less important than the binding ability of a given catalyst.

Computer Science

Parallel array characterization and preliminary SOT-MRAM modeling in MemSysExplorer

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Purpose. As processors continue to get faster, it has become more and more important that our memory systems scale to match. MemSysExplorer (MSX) is a software framework for evaluating memory systems in terms of performance, energy efficiency, and more. One critical component of MSX is the array characterizer, which predicts properties of memory chips given a high level specification. This research sought to improve the array characterizer in two ways: by improving its speed with parallelization, and by testing its ability to model arrays based on state-of-the-art SOT-MRAM cells.

Methods. For parallelization, we made use of the OpenMP library. The critical array parameter search loop, which evaluates every possible combination of array parameters to optimize for a provided target, was altered to safely distribute the work across multiple cores using a thread pool. For modeling SOT-MRAM arrays, we reviewed the literature of manufactured SOT chips, and selected the publication which provided the most empirical data. Using the array characterizer’s existing MRAM cell type as a base, a new “read pulse” parameter was added, and array characteristics were extrapolated based on the data provided in the paper for a variety of capacity configurations.

Results. The newly introduced parallelization resulted in a marked improvement in array characterization speed when using multiple cores; on the college’s FrostByte cluster, array characterization was 22 times faster on average when using 32 cores instead of just 1. Regarding the preliminary SOT-MRAM modeling, due to the novelty of the technology, there were few data points available to validate the model’s extrapolated predictions. Given identical capacity and subarray layouts to the original paper, the existing model landed within 52% of the original macro.

Conclusion. Parallel array characterization dramatically reduced search times, allowing for quicker iteration and testing. The initial work on characterizing SOT-based arrays was instructive, and as the technology develops further and more data becomes available, we will be able to further validate and refine the model.

Show, Don't Ask: Resolving ambiguities in interactive Text-to-SQL environments using differentiated databases

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Purpose. Text-to-SQL is the task of translating natural language questions into executable SQL queries, enabling non-technical users to access and analyze data. Our research studies the effects of introducing a differentiation-based approach in ambiguity resolution in Text-to-SQL systems. Specifically, it investigates under what conditions this approach improves performance on BIRD-Interact, a Text-to-SQL benchmark for evaluating LLM performance under ambiguous settings.

Methods. Given an ambiguous task from a simulated human user, a Text-to-SQL agent generates multiple candidate SQL interpretations of the user's request. The system synthesizes database instances that distinguish a pair of candidates, based on a technique known in programming as fuzzing. Provenance analysis is then conducted on the database to show the user the smallest set of relevant database and result tables. The simulated user then selects the output reflecting their intent by comparing candidate outputs against a ground-truth solution. This decision informs a candidate pruning process, guiding candidate regeneration that reflects a more informed understanding of the user's intent. Through multiple rounds of these interactions, the Text-to-SQL agent resolves the ambiguities in the user's initial prompt, converging to a final SQL that can be evaluated against a ground-truth SQL on BIRD-Interact's benchmark databases.

Results. Initial results are verified on a set of 50 questions chosen across all 22 BIRD-Interact databases. The fuzzing module achieves 98.4% recall in differentiating the candidate pairs during interactions. Overall task accuracy ranges from 8-14% on the test set, a competitive result compared to the performance of top models on BIRD-Interact.

Conclusion. Database fuzzing emerges as an effective technique to differentiate two queries, as demonstrated by the high recall (98.4%) score. But the low success rate of the full interaction loop (8-14%) highlights the challenges of the BIRD-interact tasksuite. It suggests that the optimal approach might require a combination of database differentiation with the traditional natural language ambiguity resolution. This represents the current direction of the research.

Mathematics and Statistics

An Agent-Based Model of *C. difficile* Transmission Between Hospitals and Communities

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Purpose. *Clostridioides difficile* (*C. difficile*) is a highly contagious bacteria that spreads through the fecal-oral route. *C. difficile* has typically been viewed as a nosocomial infection, but incidence has increased in the community in recent years. This model simulates the transmission of *C. difficile* between hospitals and communities to analyze the primary causes of colonizations in both the hospital and community, determine which mitigation strategies are the most effective in reducing incidence, and investigate the amount of time it takes for *C. difficile* to infiltrate the community from the hospital and vice versa.

Methods. We developed an agent-based model using NetLogo that represents 10,000 people in a community including a hospital, houses, workplaces, schools, and public restrooms. Individuals can move between four disease classes: resistant, susceptible, colonized (asymptomatic), and diseased (symptomatic). To obtain baseline results, we ran 100 copies of one-year simulations in our community. Then, we implemented various mitigation strategies to see which most decreased incidence of colonization and disease.

Results. Preliminary results show that decreasing length of stay in the hospital is the most effective mitigation strategy for reducing incidence of colonization and disease in both the community and the hospital. The model shows that in the hospital, patients are more likely to become colonized with *C. difficile* through contact with contaminated surfaces rather than interactions with contaminated people. In the community, workplace interactions are the most common cause of colonization. Lastly, the mean time required for *C. difficile* to infiltrate the community from a contaminated hospital is approximately 12 times longer than the mean time required for the reverse.

Conclusion. Our model highlights the potential impact of improving hospital efficiency on reducing the incidence of *C. difficile* not only in hospitals but also in communities. The prevalence of colonization associated with environmental surfaces in the hospital also highlights the importance of effective cleaning protocols and adherence to these protocols.

Supersymmetric Quantum Mechanics on a Simplicial Complex

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Purpose. The path integral formulation of quantum field theory, while successful and insightful, is computationally difficult to apply even in very simple cases because of its requirement that one takes integrals of infinite measure. In this paper, we extend a "toy model" of this formulation which instead uses a simplicial complex as a discrete configuration space. In past works, graph quantum field theory in the free case found supersymmetry to be conserved for all non-trivial graphs, which is a barrier to a realistic model, since we do not see a perfect mass symmetry between bosons and fermions. We expand the graph formulation to an n -dimensional simplicial complex, and then apply potential fields in order to break supersymmetry.

Methods. We define a de Rham Cohomology describing the energy of n -dimensional states. Then we prove the Morse inequalities, which state that the number of n -dimensional critical points of a Morse function is bounded from below by the number of n -dimensional holes in the simplicial complex for any Morse function. A Morse function is a function from the set of simplices to the real numbers that has a clear flow from high to low dimension, where higher dimension simplices tend to have a higher assigned values than lower dimensional ones. Then, we use the de Rham Cohomology to create Hamiltonians for each dimension, which are square positive definite matrices called the n -dimensional Laplacian. Then, we add potential fields to this Hamiltonian and search for 0-energy states. Simplicial complexes for which there are no 0-energy states require supersymmetry breaking.

Results. We find that supersymmetry is broken for almost all cases of potential fields added to graphs, excluding very specific cases. This makes graph quantum mechanics viable as a descriptive model of the real world, which exhibits supersymmetry breaking. We additionally prove the Morse inequalities on a simplicial complex, validating the supersymmetric approach to a simplicial complex.

Conclusion. Quantum field theory on a simplicial complex is a valid representation of 3D space mathematically, since supersymmetry can be broken and the Morse inequalities hold. However, more research is required to discern whether it accurately approximates continuum mechanics.

Medicine

Removing Barriers to Alcohol Use Disorder Treatment: Public Interest in Over-the-Counter Naltrexone

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Purpose. This study aims to explore public interest in over-the-counter (OTC) naltrexone and assess whether this approach may serve as a feasible strategy to expand access to alcohol use disorder treatment. Individuals' awareness, attitudes, and concerns toward OTC naltrexone have not yet been systematically studied, so this information is needed to guide future policy and education efforts to increase access to safe, low-barrier, and effective treatment for AUD.

Methods. An over-the-phone survey assessed alcohol use history, awareness of naltrexone, willingness to use naltrexone if it was available OTC, perceived benefits and concerns related to OTC use, and preferred sources of guidance if such a medication were available without a prescription. Inclusion criteria were English-speaking, 18+ adults with self-reported alcohol use at any level.

Results. The results of this retrospective study indicate statistically significant positive effects of mammoth stuffed animals on student well-being. Analysis of the quantitative data from surveys revealed a significant reduction in students' reported stress levels ($p < 0.001$) and a notable increase in self-esteem ($p < 0.05$) when interacting with these plush companions. Moreover, qualitative insights obtained from interviews and observations highlighted the presence of mammoth stuffed animals as a significant factor contributing to a sense of security and emotional support among students. The findings emphasize the potential efficacy of unconventional interventions in promoting student welfare and underscore the importance of considering innovative approaches to enhance well-being in educational settings.

Conclusion. Targeted Demand: Individuals diagnosed with AUD (84.3% aware, 65.7% likely/highly likely to use) and past naltrexone users (67.3% likely/highly likely to use, 90.4% likely/highly likely to recommend) showed the greatest interest in OTC access. Broad Acceptance: Across AUDIT-C groups and familiarity levels, over 80% anticipate a positive impact from OTC naltrexone and would recommend to others. More education needed: Concerns highlight the need for clear instructions due to lack of medical supervision in an OTC setting. In conclusion, this retrospective study provides compelling evidence that mammoth stuffed animals have a positive impact on student well-being. The presence of these endearing companions is associated with reduced stress levels, increased self-esteem, and a sense of emotional security. These findings support the integration of unconventional interventions to enhance student welfare in educational settings.

Neuroscience

The Effect of Anesthetic (Tricaine) on Larval Zebrafish Startle Habituation

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Purpose. The mechanism of general anesthesia is incompletely understood. Anesthetic effects are dose-dependent and thought to shut down more complex functions (like memory and learning) before basic functions (like reflexes). This study investigates the effect of tricaine, a common fish anesthetic, on larval zebrafish habituation, or reduced response to repeated startle stimuli. Zebrafish startle can be characterized by an extracellular electric field potential; by examining the probability and shape of this field potential between tricaine-anesthetized fish and control fish, this research asks: does low-dose tricaine selectively impair habituation before the startle reflex? More broadly, this research seeks to characterize the state of brain function in an anesthetized subject.

Methods. Zebrafish larvae were divided into control (n=13) and tricaine-anesthetized (n=10) groups. All fish were transgenically modified to express ChR-2 in hair cells, allowing them to optogenetically startle to 480nm blue light. Larvae were head mounted to a dish in low-melt agarose gel; anesthetized fish were incubated in a typical laboratory dose of 32mg/mL tricaine, while controls received a sham incubation in RODI water. After 10 minutes, two electrodes were placed by the head and tail. Larvae were given 5 baseline blue-light startles with 90-second intervals and a habituating series of 100 startles with 5-second intervals. Electric field potential data were recorded and graphed. A new bulk-analysis tool, developed for this project, was used to examine startle probability and amplitude (in mV) of each field potential.

Results. Anesthesia does not disrupt habituation and causes greater habituation and stronger startles. Anesthetized fish startled 25.6% less than control fish post-habituation ($p=0.067$). Surprisingly, anesthetized fish also had a significantly higher field potential amplitude than controls ($p<0.0001$), as well as more variability in field potential amplitude. Since field potentials are a proxy for muscle depolarization, these data mean that anesthetized fish had stronger, less precise muscle responses to startle stimuli than control fish.

Conclusion. Complex brain functions may not be shut off cleanly during anesthesia. We originally hypothesized that anesthesia would disable habituation in fish while leaving the startle reflex intact. However, the exaggerated habituation and startle intensity found in this research suggests that modulation of the startle reflex is only partially disabled by anesthesia. It is possible that inhibitory components of the startle circuit were impacted, while other activating forces remained. In human anesthesia, the sensitivity of inhibitory and excitatory neurons to anesthesia may also be variable. The way an abnormal combination of active inhibitory and excitatory neurons could impact arousal and sensory processing during anesthesia remains unclear and warrants further exploration.

The Role of m6A mRNA Methylation in D2 Striatal Neurons During Pavlovian and Instrumental Learning and Extinction

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Purpose. To understand learning, it is imperative to unravel the cellular and molecular mechanisms regulating changes in protein synthesis in neurons undergoing plasticity. The post-transcriptional methylation of adenosine into N6-methyladenosine (m6A) within mRNA transcripts is particularly efficient in modifying protein synthesis in neurons. This methylation is catalyzed by the METTL14-METTL3 heterodimer complex, and previous work has found that the expression of METTL14 in D1 and D2 medium spiny neurons in the striatum—a brain region in the basal ganglia crucial for learning—plays different roles in motor learning and drug-dependent learning. The current study investigates the cell-type dependent role of METTL14 in associative learning by examining the effect of Mettl14 deletion in D2 striatal neurons on Pavlovian and operant learning and extinction.

Methods. Five A2A-Mettl14 knockout mice and 5 control mice were used in behavioral experiments carried out in Med-PC IV operant boxes. Prior to the start of each experiment, mice were food deprived for one day. For Pavlovian learning, mice first underwent 13 days of acquisition, during which they experienced a signal light followed by a sugar pellet reward. They subsequently underwent 5 days of extinction, during which the reward was removed, and 1 day of probe, during which the reward was restored. For operant conditioning, mice underwent 7 days of acquisition, during which they lever pressed for sugar pellets on a Fixed Ratio 5 schedule. This was followed by 5 days of extinction and 1 day of probe. All data analysis was performed using GraphPad Prism.

Results. No significant overall difference in head entries was observed between control and knockout mice during Pavlovian acquisition, extinction, or probe. Notably, however, knockout mice showed significantly higher baseline head entries than controls specifically on days 5 and 6 of acquisition ($p < 0.05$). No significant difference in lever presses was observed between control and knockout mice during operant acquisition or probe; however, knockout mice showed significantly lower lever presses on the first day of extinction compared to controls ($p < 0.01$), though this difference did not last for the rest of the extinction period.

Conclusion. In general, the expression of METTL14 in D2 striatal neurons does not seem to play an important role in associative learning or extinction in mice. However, METTL14 expression in D2 striatal neurons could potentially be involved in the early refinement of behavioral patterns during associative learning, as suggested by heightened baseline head entries exhibited by knockout mice during Pavlovian acquisition.

Physics and Astronomy

3D Printing Metastructure Legs for High-Efficiency Thermoelectric Generators

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Purpose. Global energy use is rising unsustainably, with excessive energy being lost as wasted heat in settings like data centers, automobile exhaust, and power plants. Thermoelectric generators (TEGs) make electricity from heat, and could recover this wasted energy on a large scale. However, current state-of-the art models suffer from high manufacturing costs and low efficiency. The goal of this project is to develop an optimal procedure and formulation to 3D print high-solid-loading (90 weight%) Bi_{0.5}Sb_{1.5}Te₃ metastructure legs for use in a high-efficiency thermoelectric generator using direct ink writing (DIW).

Methods. Different combinations of organic binders were tested along with pre and post processing procedures to create the most printable slurry that maintained a high efficiency.

Results. Multiple slurries were created that could print through an improved nozzle size of .60 mm down from .84 mm. These slurries also achieved ~90 weight% loading. Sieving, ball milling, and planetary mixing proved to be effective at improving the printability of the slurries. Nozzle clogging remained the dominant problem for longer print times. A thermoelectric test yielded a figure of merit of .62 which is competitive with top of the field formulations.

Conclusion. This study demonstrates that direct ink writing (DIW) 3d printing has the potential to manufacture complex thermoelectric generators with high loading thermoelectric materials. The thermoelectric performance of these materials has been improved along with advances in printability. Their high efficiency provides a new direction for sustainable energy solutions.

A Pneumatically Actuated Optical Mount for Enabling Beam Interchange

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Purpose. We imprint non-trivial topologies on trapped Rubidium-87 Bose-Einstein condensate samples using selected atomic transitions driven by laser beams. To refine our transition-driving ability, we are incorporating an improved projection beam; however, this setup relies on multiple distinct beams propagating to the atoms, in cycles 30 seconds apart, through different optics along the same stretch of beam path. To make beam interchange possible, this project aimed to develop a computer-controlled, translating optical mount.

Methods. To move the optic, we preferred a pneumatic cylinder to alternatives like piezo actuators—which could introduce excess acoustic noise—and commercial flipper mounts, which do not fit within apparatus geometry. Hence, motion is propelled by an SMC 2-inch stroke cylinder and confined by four custom-machined parts in concert with a commercial guide rail. Air flow to the cylinder is controlled by a 3/2 solenoid valve, which takes in a high or low signal from a combined digital and analog soldered circuit. This circuit allows for manual or computer control. Through this system, the optical mount translates a mirror into apparatus beam path, directing one magneto-optical trap (MOT) beam into the Rb-87 cell before moving it out of beam path so that projection and imaging beams can proceed unobstructed.

Results. Relief cuts and kinematic stops ensure the mount's movement is stable. The soldered circuit gives computer control and, after adaptation, controls coolant pumps on the optical table. Further, the custom parts have several tapped holes which interface with the optics, providing adaptability as we prepare the mount for experiments.

Conclusion. This well-constrained, computer-controlled optical mount helps us meet the challenge posed by beam interchange. The mirror attached to the cylinder system directs beams to the atoms as intended, and computer control allows the mount's timing to align with the cycles of lasers in our experiment. Currently, the cylinder needs to be designed into the space near the Rb-87 cell. Together with our ongoing work to make beam paths more compact, this will help us switch to using a Spatial Light Modulator (SLM) to drive atomic transitions. Patterned light from the SLM will further enable us to imprint different excitations on the condensates and study their interactions.

Brane trajectories in cosmological spacetimes

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Purpose. A fundamental mystery of our universe is its particular rate of expansion, fueled by an outward pressure known as dark energy. While we can observe the impacts of dark energy on our universe, we cannot experimentally compare different universes to determine the causes or consequences of a specific dark energy content. Instead, we use theoretical methods to better understand the role of dark energy in cosmology through describing the behavior of objects in 3+1-dimensional spacetimes.

Methods. We apply mathematical tools from general relativity, differential geometry, and string theory to study the trajectories of general-dimensional objects moving in spacetimes, known as branes. By applying the variational principle to brane area-action, we derive the equation of motion for a general brane which is dependent on its geometry. We then apply this formalism to compare the trajectories of massive, spherical 2-branes in Minkowski and de Sitter spacetimes.

Results. The equations of motion for a massive, spherical 2-brane in Minkowski spacetime demonstrate that it tends toward shrinking, regardless of initial conditions. By contrast, a massive, spherical 2-brane in de Sitter spacetime has a critical radius at which the inward surface tension and outward pressure perfectly balance. When the brane is larger than the critical radius, it tends toward expanding as the outward force due to dark energy is greater than inward force from surface tension. Correspondingly, when the brane is smaller than the critical radius, it tends toward shrinking. These results align with their classical analogue, where higher pressure within a bubble allows it to hold its form, while no pressure difference would lead to a shrinking bubble.

Conclusion. While the dark energy content of a spacetime is conventionally seen as a constant, branes with negative energy can be absorbed by black holes, leading to a change in the dark energy content of the spacetime. This research opens the door to further study of mechanisms for changing the dark energy content of spacetimes via black hole energy extraction. Our eventual goal is to describe negative-energy charged branes absorbed by a rotating Bañados-Teitelboim-Zanelli (BTZ) black hole, as a model mechanism enabling the fine-tuning of our universe's dark energy content.

Modelling the Twins Detector at the Black Hills Underground Campus (BHUC)

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Purpose. During this research project, a simulation was created to model the Twins detector in the Sanford Underground Research Facility's (SURF) Black Hills Underground Campus (BHUC). The Twins is a dual crystal High Purity Germanium (HPGe) detector used for low background counting. Low background counting is a method for detecting the amount of radiation coming from a given sample as elements in the sample decay. The focus of this project was to use Geant4, a particle physics simulation software, to model these interactions and compare them with the Twins' data. Creating a simulation and comparing its results with the Twins' data is an important pursuit because scientists can measure how efficient the Twins detector is at collecting data, since the simulation allows us to specify how many gamma rays are initially shot vs. how many are detected, which is not a measurement the Twins detector is capable of.

Methods. We created a Geant4 simulation which mimicked the Twins detector's geometry, created a mock sample to shoot gammas at the crystals, and recorded the results as energy spectra. Energy spectra are graphs of the energy the gamma rays deposit in the crystals as they pass through; this energy is what creates the electrical current scientists record as a signal. These spectra are compared with the spectra the Twins detector produces.

Results. So far, the simulation has successfully mimicked the Twins detector, but we made simplifying assumptions to make the simulation, which impacts how closely it resembles the Twins detector. For example, we simplified the geometry of the detector into cube shapes, despite the real Twins detector's crystals being u-shaped. Additionally, the simulation does not include the error in measurements taken in the real world. For example, the electronics the Twins detector uses to collect data "smear" that data due to limits on how quickly the electronics can identify electrical signals. Therefore, while the simulation does recreate many of the same processes as the Twins detector, they do not currently match well enough to compare.

Conclusion. The simulation is still too ideal to be compared with the Twins detector, so this project's next steps are to simulate the real world's noise, such as the error the electronics of the Twins detector introduces to its data. Eventually, the goal is to measure the efficiency of the Twins detector at low background counting.

Protoplanetary Disk Processing: Centering and Deprojecting

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Purpose. Just after a star is formed, it is surrounded by a spiraling clump of dust and gas that flattens into a disk called a protoplanetary disk. These disks are the birthplace of planets and have been observed around nearby stars using millimeter and near-infrared light. Images show that most protoplanetary disks have features such as spirals or rings, which could signal planet formation. However, these are not the only theoretical causes of such features: temperature, magnetic fields, or other dynamic processes could also contribute. To better understand what affects these disk parameters, there needs to be a normalized method for classifying and analyzing disk substructures.

Methods. Here, the Comparative Analysis Tool for Normalized Imagery and Profiles, or CATNIP, comes in. This computational tool aims to create a more unified classification of structures within protoplanetary disks by using azimuthal and radial profiles. This project added a Python program to CATNIP that estimates the center and inclination of the disk. To find the center, two key assumptions were made: the disk images should be circular when face-on and mostly azimuthally symmetric. Using these assumptions and a contour ellipse-fitting program, the tool deprojected the image using each ellipse and assigned it a score for how well it aligned with the above assumptions. Then the ellipse with the best score returned the estimated center and inclination of the disk.

Results. The program successfully identified 87% of the disks' centers accurately, and found an accurate deprojection 81% of the time. These percentages remained roughly consistent across all substructures, as well as for near-infrared and millimeter data. This implies tool performance is not reliant on the substructure present in the disk.

Conclusion. In the future, this tool should be integrated into the CATNIP user interface. The difference between an accurate center and deprojection significantly changes the data collected from azimuthal and radial profiles. Then the CATNIP tool can be used to compare disk parameters across the documented population and find quantitative patterns for what is causing these substructures.

SnO₂-Coated AFM Tips for Nanoscale Electrical Characterization of Perovskite Solar Cells

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Purpose. Currently, commercial solar devices do not exceed 30% power conversion efficiency (PCE). Perovskites present a valuable alternative to traditional silicon solar cells. The tunable bandgap, energy conversion efficiency, and quick energy payback of perovskites are inherent to their unique crystalline structure. Improving perovskite solar cell technology requires direct analysis of the perovskite layer at the Electron Transport Layer (ETL) interface. This interface is analyzed using an Atomic Force Microscope (AFM), which moves a sharp probe tip across a surface to map topography and local conductivity at the nanoscale. This research aims to improve perovskite analysis and effectively increase PCE, actualizing solar's potential in the renewable energy landscape.

Methods. Various methods were explored for depositing SnO₂, a common perovskite ETL, onto microscopic AFM probe tips. Film thickness, stoichiometry, and electron mobility were optimized. Although not viable for the AFM probe itself, spin coating is a standardized technique that was used as a target for feasible tip deposition methods. Substrates were coated to establish parameters that were then applied to AFM tips. Tested methods included sputtering, where deposition time and oxygen concentration were optimized, and dip coating, where withdrawal rate and solution concentration were adjusted. Thin films were characterized using AFM and UV-vis spectroscopy. Additionally, bare tips on coated substrates were compared to coated tips on bare substrates.

Results. Analysis revealed that increasing oxygen concentration during sputtering yielded a bandgap that approached the spin coated thin film. Meanwhile, dip coated substrates exhibited bandgaps and electrical characteristics more closely aligned with spin coating. Electrical characterization of SnO₂-coated AFM tips showed that ideal substrate deposition conditions are not directly transferable to ideal deposition conditions for tips. This mismatch in electrical behavior is likely tip films are significantly thinner than substrates films. Ultimately, the sputtered tips exhibited more uniform coatings than the dip coated ones.

Conclusion. Although dip coating naturally yields a bandgap and electrical characteristics closer to spin coating, sputtering has greater potential due to its tunability and also results in a compact layer that has not yet been achieved through dip coating. Future work will optimize sputtering for improved electrical properties, refine dip coating for sharp tip geometry, and compare coated versus bare tips on perovskite. These findings provide a foundation for advancing nanoscale characterization and realizing high-efficiency perovskite solar cells.

The Effect of Outer Giant Planets on Inner System Architectures

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Purpose. Understanding the processes that dictate planetary formation and evolution is crucial to constraining how the architectures of exoplanetary systems emerge, as well as for determining regimes of stability and chaos. Observations from NASA’s Kepler mission revealed a remarkable trend of intra-system uniformity in inner terrestrial planet masses, radii, and orbital spacings, a phenomenon referred to as “peas in a pod” architectures. Evidence suggests that these system architectures may result from later dynamical sculpting, rather than being a natural outcome of early planet formation. Furthermore, systems featuring an orbital gap between two of the inner compact planets preferentially harbor an outer gas giant, suggesting a possible dynamical process whereby the outer giant induces instabilities, collisions, and ejections that reshape inner system architectures.

Methods. Here, we investigate whether the inward migration of an outer giant planet can sweep strong enough gravitational resonances to adequately pump the eccentricities and inclinations of inner planets to generate the observed gaps. Using the N-body integrator REBOUND, we simulate randomized inner compact terrestrial systems and track their evolution as a Jupiter-mass planet migrates inward. We complement these numerical simulations with analytical Laplace-Lagrange secular theory calculations using celmech to evaluate mean motion resonances and secular resonance crossings over time.

Results. Preliminary results demonstrate that a single outer giant is too weakly coupled to the terrestrial inner planets to sweep strong secular resonances. Moreover, significant differences in semi-major axes between the outermost terrestrial planet and the giant planet ensure that only high-order mean motion resonances are encountered during migration, which are too weak to trigger dynamical sculpting.

Conclusion. Thus, our early results suggest that a second giant planet may be required to sweep resonances more strongly, or that another unaccounted-for dynamical process is shaping the system architectures we observe.

Water-Cooling Acousto-Optics for a Paul Ion Trap

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Purpose. Our lab uses a linear Paul trap to hold atomic and molecular ions for precision spectroscopy. Laser-cooled calcium ions are used to cool and detect trapped ions, requiring precise control of the frequencies of several laser beams. Acousto-optic modulators (AOMs) drive radio-frequency (RF) signals into crystals, producing acoustic waves that can diffract and manipulate the frequency of laser light. However, the RF power dissipated in an AOM generates heat, which can produce temperature gradients and affect the stability of the diffracted beam. So, we constructed a water-cooling system to thermally stabilize the AOMs in our laser system.

Methods. We adapted a commercial computer water-cooling system for the AOMs. The system consisted of a water reservoir, pump, radiator and fans connected through flexible tubing. Two water-cooling blocks would sandwich each AOM. The two AOMs, one resonant and one detuned, were connected in series in one water loop. We monitored the AOM temperatures and subsequently characterized the optical system by measuring laser power, photodiode voltage, diffraction efficiency, and beam alignment.

Results. Without active cooling, the AOM temperature increased to approximately 40 C during operation. With water cooling, the measured AOM temperature remained at room temperature, indicating that the system effectively removed any heat we could measure externally. After more realignment, the resonant AOM produced first-pass and second-pass diffraction efficiencies of approximately 74% and 62%, respectively. Although the external AOM temperatures were successfully stabilized, the measurements did not directly determine the temperature distribution within the AOM crystals.

Conclusion. The water-cooling system successfully prevented the large temperature increase previously observed on the AOMs and maintained both AOMs near room temperature during operation. However, because the cooling blocks contact the exterior of the AOM rather than the crystal directly, further measurements are required to determine whether the system also suppresses the internal temperature gradients.

Psychology

Community Perspectives on Civic Engagement: Findings from Amherst's Civic Academy

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Purpose. Civic engagement can take many forms, including voting, volunteering, attending town meetings, and working with others to address local concerns. This study looks at how Amherst residents understand their role in the community and whether participating in the Town of Amherst's Civic Academy, an 8-week, in-person introduction to town government, affects how connected they feel to Amherst, how much influence they believe they have, and how they view local government.

Methods. We interviewed 28 Amherst residents: 10 Civic Academy participants, 8 residents who applied but were waitlisted, and 10 other community members with similar backgrounds. Interviews lasted between 45 minutes and two hours. Prompted by a semi-structured interview guide, residents talked about their connection to Amherst, the issues that matter to them, and the ways they participate in the community. They also completed graphic elicitation activities where they identified important issues, their advocacy network, and a timeline of their civic identity development. Thematic analyses, which are ongoing, involved all members of the research team independently reviewing the interviews and meeting to group similar responses into shared themes.

Results. Our early findings show four common themes in how residents talked about civic engagement. The Institutional theme describes how government, money, and other larger systems affect residents' ability to participate. The Inter/generational theme shows how family, education, and experiences across generations shape community involvement. The Team/work theme focuses on the importance of working with others, as well as the challenges that can make teamwork difficult. Finally, the Co/compound theme describes how residents combine different approaches, such as voting, building relationships, sharing information, and joining community groups, to make a difference. Across these themes, residents often felt they had the least influence over the issues they considered most important.

Conclusion. These early findings show that civic engagement is about more than voting or attending town meetings. How people participate is also connected to their relationships, life experiences, access to resources, and belief that they can make a difference. As we continue our analysis, we will compare the three groups to better understand how the Civic Academy may shape residents' connection to Amherst, confidence in their ability to create change, and views of local government. Through our ongoing research-practice partnership, these findings may also help guide future Civic Academy programs.

Sixty Years of Ethnic Studies: A Scoping Review of Emerging Research on Ethnic Studies Policy and Impacts on Adolescent Development

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Purpose. Amidst response to student protests for an education that reflects the experiences and perspectives of communities of color, ethnic studies was created to examine our nation's diverse racial and ethnic groups through social, economic, and historical lenses (Sleeter & Zavala, 2020, p. iv). But in recent years, amidst social justice demonstrations and divided legislative proposals to expand or ban ethnic studies, it is important, now more than ever, to understand the implications of ethnic studies for adolescent development. Our scoping review will address the following questions: How are schools defining and implementing ethnic studies? What mechanisms have been identified as the "active ingredients" of ethnic studies? And what has research shown about the implications of ethnic studies for youth's development? With advances in both theoretical and empirical research, we aim to provide an updated review of ethnic studies in U.S. middle and high schools to inform policy and practice.

Methods. Our review followed PRISMA guidelines for scoping reviews (Tricco et al., 2018) to synthesize the most recent trends in ethnic studies implementation and research. Results were imported into Covidence software for automatic deduplication, screening, and data extraction. All titles and abstracts (n=4236) and full texts (n=132) were screened by two of the four research assistants using a criteria list, achieving high group agreement (>95%). ProQuest was used to extract works from two databases (PsycINFO and ERIC) from January 2019 to March 2026.

Results. Ethnic Studies classes can bolster success across a range of developmental tasks in a diverse society. As we continue data extraction, with planned completion in Fall 2026, there are preliminary results to highlight. More articles on ethnic studies have been published in the past six years than in the previous 40 years (1980s–2020) covered in Sleeter & Zavala's (2020) scoping review. And just as the field of ethnic studies continues to expand, so do the research methods being used to understand it, ranging from large-scale quantitative analysis to autoethnographic research by ethnic studies teachers themselves. And as the U.S. K–12 student body continues to diversify, teachers are reporting innovative pedagogies they use to integrate ethnic studies in the classroom.

Conclusion. Almost 60 years ago, student protestors at San Francisco State College sparked a new era for U.S. education, one focused on implementing the experiences, voices, and perspectives of those who have been marginalized in U.S. history. As we synthesize the results of our research, we plan to submit our scoping review to the Special Issue in the Journal of Research on Adolescence in February 2027. And with our results, we hope to translate our findings for others through knowledge translation, including school officials, teachers, policymakers, parents, community members, and, most importantly, those who helped make what we now know as the field of Ethnic Studies: students.